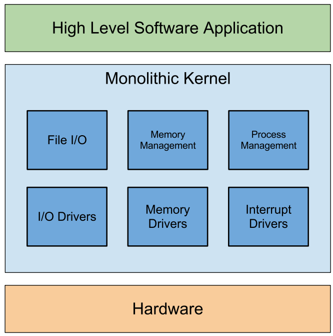
various hardware and software resources and ensure safe and reliable functioning of the system.

Embedded OSes come in three flavours: monolithic, layered and microkernel. These models differ based on the internals of the OS's kernel along with the additional system software incorporated in the OS.

Monolithic

In case of a monolithic OS, the middleware and the device drivers are included in the kernel itself. In most cases this OS is usually a single executable file which contains the above mentioned components. Monolithic OSes are in most cases more difficult to scale, maintain and debug than their architectural counterparts.



Monolithic OS

Usually, a more popular alternative to this type of OS is used, it is called the modularised monolithic model. In this case instead of one huge file, the OS is broken down into separate files called modules. Each module is responsible for one of the OS's many functionalities. In this case only the modules needed are loaded into the memory and it also makes scaling, maintenance, modification and debugging easy.

Linux is a popular modularised monolithic non-embedded OS. Jbed RTOS, µC/OS-II and PDOS are examples of embedded monolithic OSes.

Layered

In a layered system, the OS is divided into hierarchical layers, where the upper layers are dependent on functionality provided by the lower layer. Like a monolithic OS, this system is made available as a single, large executable file which includes the device driver and the middleware along with the kernel. But due to it's layered nature it's easier to manage and debug such systems. One of the drawback of this design is the overhead incurred due to the overhead added by the API provided at each layer.

DOS-C(FreeDOS), DOS/eRTOS, and VRTX are all examples of a layered OS.

Microkernel

This is the most popular OS model among the off-the-shelf embedded OSes. Often called the client-server OS – the main OS is stripped down to the min-